STAFF REVIEW · H446 253 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. The instruction's contents from memory towards the CPU.

2. PC.

3. One operation cannot proceed until another result is ready.

4. Q=0. The flip-flop samples D at the rising edge (D=0 at that instant); an earlier pulse between clock edges is not stored.

I Do — source worked steps

1. Predict first — Cycle 1 fetch I1. In cycle 2 decode I1 while fetching I2.

2. Explain and check — Cycle 3 executes I1 and decodes I2; I2 executes in cycle 4.

We Do — guided CHECK

1. Predict first
Begin a new fetch on each clock cycle, while older instructions advance to Decode or Execute.
Explain and check
Instruction four starts Fetch in cycle 4, Decodes in 5 and Executes in 6.

You Do — source mark schemes

1. [4] Higher clock rate may allow more cycles each second [2]; larger/faster cache can reduce average memory latency [2]; alternatively cores can run independent tasks.

2. [6] C1 I1:F [1]; C2 I1:D, I2:F [1]; C3 I1:E, I2:D, I3:F [2]; C4 I2:E, I3:D [1]; C5 I3:E [1].

3. [6] Memory stalls/cache misses [1], non-parallel parts of software [1], instructions per cycle vary [1], branch hazards can flush a pipeline [1], data dependencies stall instructions [1]; measured speedup depends on the bottleneck [1].

Exit Ticket — source answer

1. False. It usually overlaps stages to improve throughput, but each instruction still passes through its stages.